Question 1: Consider the basic SIR ODE model without demography with frequency-dependent mixing:

$$\frac{dS}{dt} = -\frac{\beta SI}{N}, \quad \frac{dI}{dt} = \frac{\beta SI}{N} - \gamma I, \quad \frac{dR}{dt} = \gamma I, \tag{1}$$

with parameters $\beta = 2/14$ infections per day, $\gamma = 1/14$ removals per day, $N = 5500$ and $\iota = 10$. We assume an initial state of $(S(0), I(0), R(0)) = (N - \iota, \iota, 0)$.
Q1a): With the setup above, the number of infectives over time is shown below:

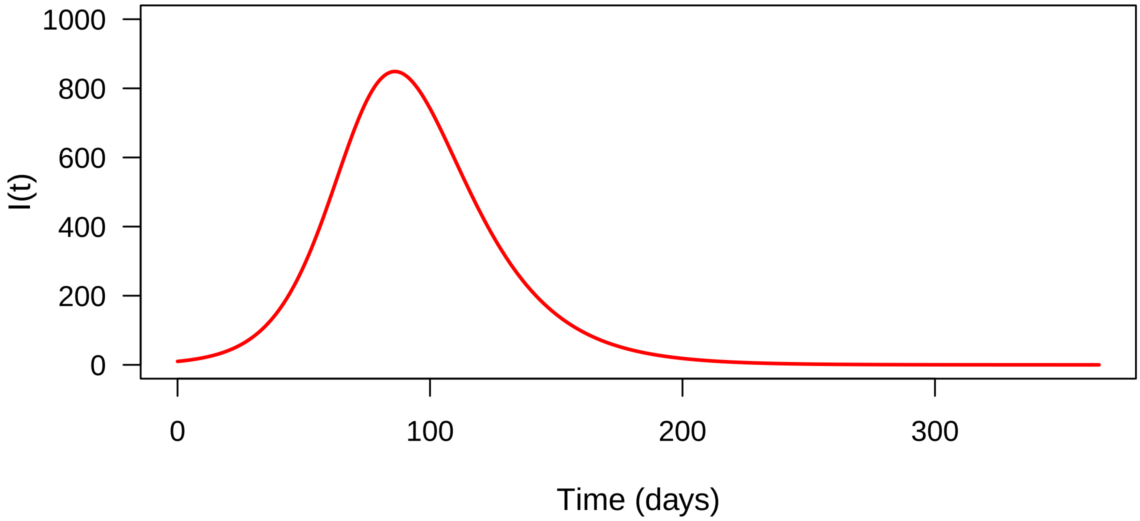

Figure 1: Number of infectives over time with parameters $\beta = 2/14$ infections per day, $\gamma = 1/14$ removals per day, $N = 5500$ and $\iota = 10$.

Q1b): We assume that the incidence of hospitalisation from a week ending on day t is $Y_t \sim \text{Poisson}(\delta I(t))$. After coding an R function to calculate the log-likelihood of the RSV data as a function of the parameters $\bar{\theta} = (\beta, \gamma, \iota, \delta)$, we evaluated it at our previous parameter choices with $\delta = 0.05$, yielding a raw output of `5537.94`. Since the `optim` function minimises by default, but we aim to maximise, our function negates this value for optimisation purposes, meaning the true log-likelihood output is `-5537.94`.

Within our log-likelihood function, we use the `dpois` function, which computes the probability density function (PDF) of the Poisson distribution given a rate parameter λ and observed data. Additionally, by setting the `log=TRUE` flag, we transform the likelihood function into a log-likelihood.

Q1c): The parameter δ represents the hospitalisation rate due to RSV. Given that Y_t denotes the weekly number of hospitalisations and follows a Poisson distribution, $Y_t \sim \text{Poisson}(\delta \cdot I(t))$, it follows that the expected number of hospitalisations per week is $\delta \cdot I(t)$. Therefore, the probability of an infected individual being hospitalised is δ. Assuming England has a population of $55,000,000$ and that we use the parameters specified before, our infection dynamics looks like:

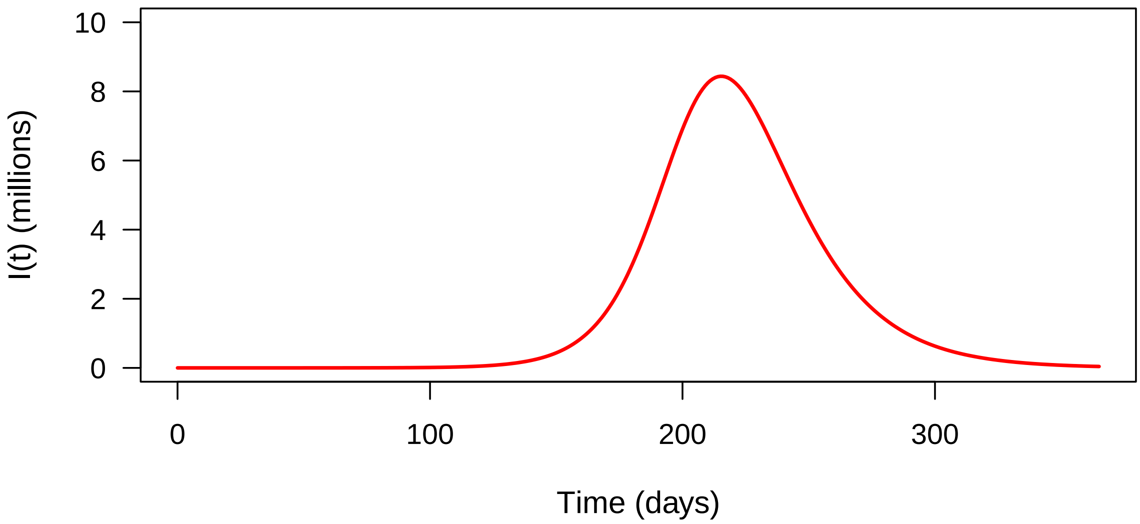

Figure 2: Number of infectives over time with parameters $\beta = 2/14$ infections per day, $\gamma = 1/14$ removals per day, $N = 55,000,000$ and $\iota = 10$.

The reason for the shift in peak infections is because we leave $\iota = 10$ so it takes longer for the peak to be achieved with a larger population size. However, we observe that the peak is approximately $8,000,000$, the same proportion when using $N = 5500$. Therefore, with $\delta = 0.05$, the expected number of hospitalisations at the peak of the epidemic is $\delta I(t_{\max}) = 0.05 \cdot 8,000,000 = 400,000$.

Q1d): Using the R function `optim`, we can determine the maximum likelihood estimator (MLE) for the parameter vector $\bar{\theta} = (\beta, \gamma, \iota)$ while keeping δ fixed at 0.05. We initialise the parameter vector $\bar{\theta} = (2/14, 1/14, 10)$ as before. Then, after running the code, we obtain our MLE as

β	γ	ι
0.08574	0.00692	77.73077

Table 1: MLE for parameter vector for (β, γ, ι).

The log-likelihood value at the MLE is `132.23`.

Q1e): We can predict the expected number of hospitalised cases (denoted as *model hospitalisations*) using our model combined with our MLE. We plot this against the observed number of hospitalised cases (denoted as *actual hospitalisations*) below:

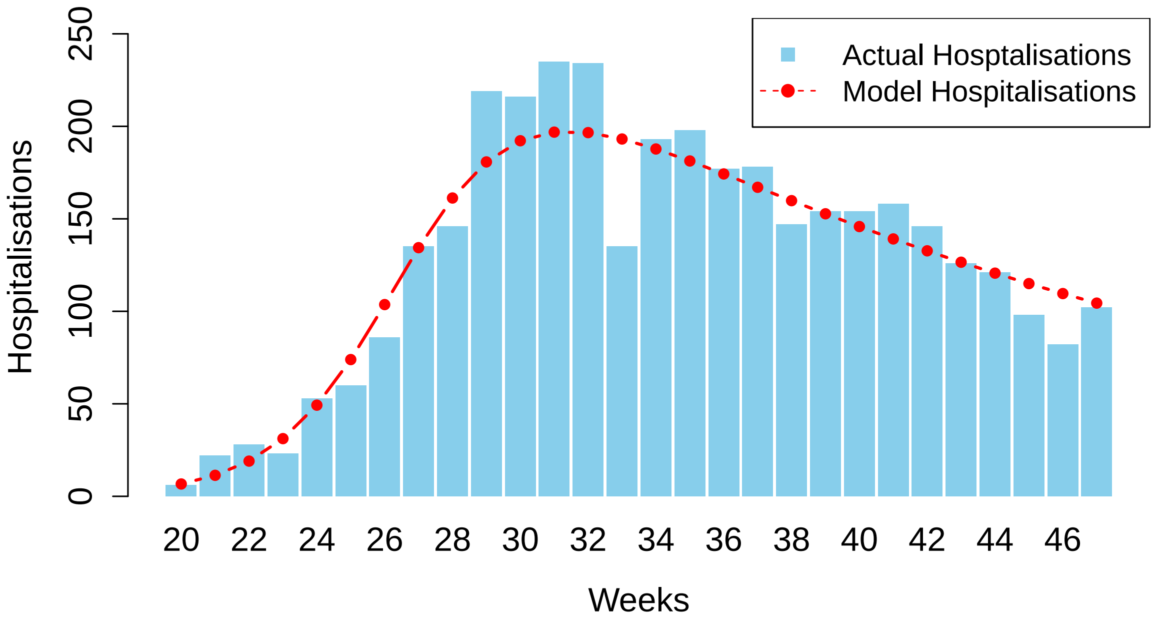

Figure 3: Fitted model predicted hospitalisations (red) and actual hospitalisations (blue) across epidemiological weeks 20 - 47 for RSV in England.

The fitted model successfully captures the overall trend of RSV. We observe a sharp increase during epidemiological weeks 20–30, which is correctly predicted by our model, followed by a gradual decrease in hospitalisations until epidemiological week 47, which is also well captured.

However, the model struggles to capture short-term fluctuations, such as the dips observed around epidemiological weeks 32, 38, and 45. This is because the fit is optimised to account for the entire dataset with equal weight, forcing it to align with the overall pattern. Additionally, the model lacks a stochastic component to capture short-term variations and noise.

Using R, we compute the R^2 score for our fitted model, which yields 0.903, suggesting that our fit is highly successful in capturing the general trend.

Q1f): To determine whether the optimiser converged correctly, we check the `$convergence` flag in the object returned by the `optim` function. Upon doing so, we obtain a response code of `0`, indicating successful convergence.

We compute the hessian matrix from the optimisation in R and we yield the following matrix:

$$\begin{pmatrix} 463988.36 & 5446.57 & 116.79 \\ 5446.57 & 18294320 & 30.88 \\ 116.79 & 30.88 & 0.035 \end{pmatrix}.$$

The Hessian matrix represents the curvature of the local minimum obtained by the `optim` function. Indeed, we observe the second order partial derivatives for each parameter with respect to the other parameters are positive suggesting that in all directions the gradient is positive. This implies that the point at which the `optim` function stopped at is a local minimum on the likelihood functions surface suggesting that the minimisation problem converged. Since, our goal was to maximise the log-likelihood function, the shape of our likelihood function will be a the inverse of the shape produced by the `optim` function so we have a concave surface around the MLE. After altering the initial values for θ, we observe slight differences in the converged solutions. This suggests that the likelihood function has multiple local maxima and that the optimiser tends to converge to the local maximum closest to the initial conditions. Additionally, one can compute the eigenvalues of the matrix to determine concavity. This is because the eigenvalues provide the diagonalised form of our Hessian matrix and we know the curvature data is preserved under this transformation and then if the matrix is positive-definite i.e. all the eigenvalues are positive, we have the rates in all direction are increasing and thus the `optim` function converged to a local minimum i.e a local maximum on our log-likelihood function surface. Using R, our eigenvalues are `18294330, 463986.7, 0.006` which confirms convergence.

Question 2: In this question, we adopt a Bayesian framework for analysis.
Q2a): The given prior beliefs are that:

$$\begin{aligned}\beta \mid \gamma &\sim \text{Gamma}(32, \gamma/20),\\ \gamma &\sim \text{InverseGamma}(6, 1),\\ \iota &\sim \text{Exp}(0.01),\\ \delta &\sim \text{Exp}(20).\end{aligned}$$

Then, the joint prior density function for parameters β and γ is given by

$$p(\beta, \gamma) = p(\beta \mid \gamma)p(\gamma)$$

since β is conditioned on γ. Assuming the other two parameters are independent from β and γ, the joint prior density function for all four parameters is given by

$$\begin{aligned}p(\beta, \gamma, \iota, \delta) &= p(\beta \mid \gamma)p(\gamma)p(\iota)p(\delta)\\ &= \frac{1}{\Gamma(32)}\left(\frac{20}{\gamma}\right)^{32}\beta^{31}e^{-\beta\left(\frac{20}{\gamma}\right)} \cdot \frac{1^6}{\Gamma(6)}\gamma^{-7}e^{-\frac{1}{\gamma}} \cdot 0.01e^{-0.01\iota} \cdot 20e^{-20\delta}.\end{aligned}$$

Then, taking logarithms to both yields yields

$$\begin{aligned}\log(p(\beta, \gamma, \iota, \delta)) = &-\log(\Gamma(32)) + -32\log\left(\frac{\gamma}{20}\right) + 31\log(\beta) - \beta\left(\frac{20}{\gamma}\right)\\ &+ 6\log(1) - \log(\Gamma(6)) - 7\log(\gamma) - \frac{1}{\gamma}\\ &+ \log(0.01) - 0.01\iota\\ &+ \log(20) - 20\delta.\end{aligned}$$

Evaluating our R function at $\bar{\theta} = (2/14, 1/14, 10, 0.05)$ yields a value of `-1.1257`.

Q2b): According the the *Moore et al.* paper on RSV, an average infectious period of 10 days was used to be consistent with previous literature. They then used values

$7-12$ for this parameter to determine the effect on the model. From experimenting, we find that a Gamma$(10,1)$ has most of its mass between $7-12$ and a mean of 10. Since the mean infectious period is $1/\gamma$, that leaves an InverseGamma$(10,1)$ prior for γ. We know that R_0 for RSV is typically around 3.0 and ranges in the interval $[1,5]$. Since a Gamma$(10,1/3.3)$ has most of the mass between $[1,5]$ with a mean of approximately 3, our prior for $\beta \mid \gamma$ should be $\beta \mid \gamma \sim$ Gamma$(10,\gamma/3.3)$. We know $\delta \in [0,1]$ as a probability, therefore $\delta \sim$ Beta$(3,55)$, which has a mean near 0.05, is a suitable prior distribution. For ι, we assume the initial infections is 200 so we need a rate parameter λ such that $1/\lambda = 200$, hence $\iota \sim$ Exp(0.005).

Summarising, our prior distribution for the parameters is given as

$$\begin{aligned} \beta \mid \gamma &\sim \text{Gamma}(10, \gamma/3.3), \\ \gamma &\sim \text{InverseGamma}(10, 1), \\ \iota &\sim \text{Exp}(0.005), \\ \delta &\sim \text{Beta}(3, 55). \end{aligned}$$

Q2c): Using an adaptive random walk Metropolis algorithm, we fit our model to the incidence data within a Bayesian framework using the prior distribution given in part 2A. We initialise the chain from $\bar{\theta} = (\beta, \gamma, \iota, \delta) = (0.07, 0.03, 80, 0.15)$. The posterior distribution for the parameters $\bar{\theta}$ is given in the following table:

	β	γ	ι	δ	R_0
Median	0.0708	0.0269	88.760	0.155	2.624
2.5%	0.0670	0.0216	71.454	0.121	2.193
97.5%	0.0753	0.0338	108.867	0.207	3.219

Table 2: Posterior distribution for the parameters $\bar{\theta} = (\beta, \gamma, \iota, \delta)$ and R_0 with the posterior median and equal-tailed 95% marginal credible intervals for each parameter and R_0.

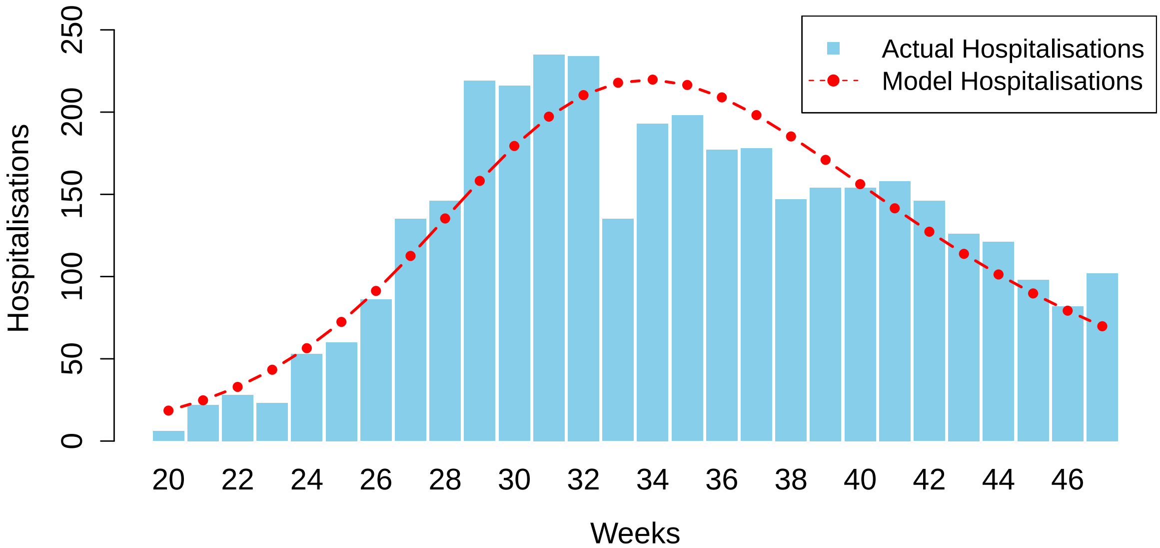

Figure 4: Fitted model predicted hospitalisations (red) and actual hospitalisations (blue) across epidemiological weeks 20 - 47 for RSV in England using adaptive MCMC algorithm.

Q2d): The following graphs presents trace plots for each parameter which includes the acceptance rate.

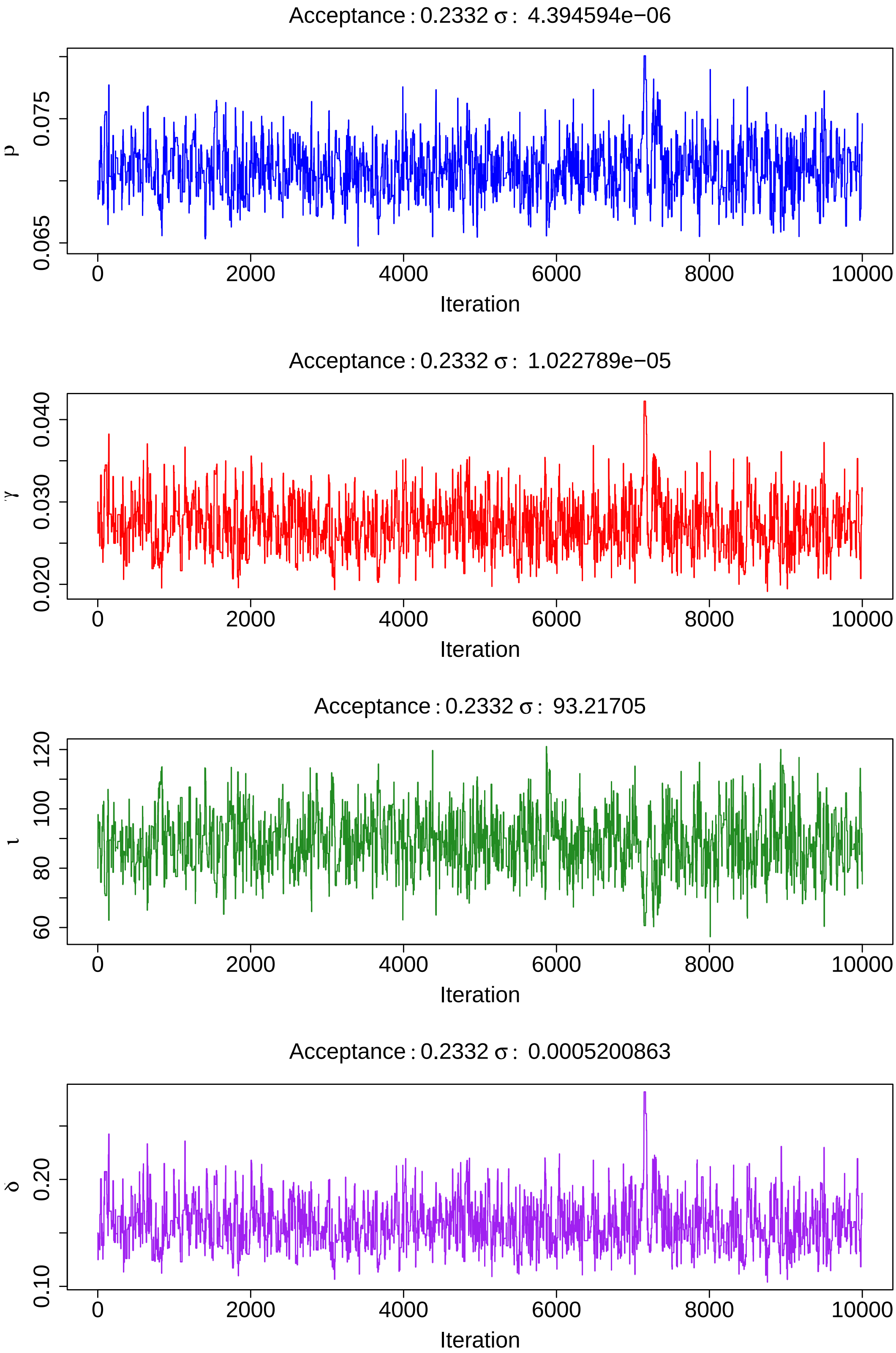

Figure 5: Trace plots.

The trace plots suggest that our initial value for `Sigma`, the variance-covariance matrix for the parameters, is a good estimate, allowing the MCMC algorithm to mix well early on. For this reason, I chose to burn 10% of the iterations when computing our posterior distribution metrics as a precautionary measure to ensure the MCMC algorithm has fully entered the posterior parameter space.

The fluctuation of each trace plot around a stable mean indicates good mixing. Additionally, our acceptance rate is 23.32%, which falls within the recommended 20% – 40% range. In fact, the asymptotic optimal acceptance rate is 23.4%, suggesting that the mixing is near-optimal in our simulation.

We run three separate chains with initial parameter values $(0.07, 0.03, 80, 0.15)$, $(0.06, 0.025, 85, 0.14)$, and $(0.08, 0.035, 75, 0.16)$ to compute the *Gelman-Rubin* diagnostic metric and the effective sample size, providing a more robust assessment of convergence. For the *Gelman-Rubin* diagnostic, we obtain the following results:

	Point est.	**Upper CI**
β	1	1
γ	1	1
ι	1	1
δ	1	1

Table 3: Gelman-Rubin diagnostic for 3 MCMC chains at initial parameters $(0.07, 0.03, 80, 0.15)$, $(0.06, 0.025, 85, 0.14)$ and $(0.08, 0.035, 75, 0.16)$.

When the upper limit of the *Gelman-Rubin* diagnostic is close to 1, it indicates convergence, suggesting good mixing and stability in our simulation.

The effective sample size (ESS) measures how much independent information our MCMC chains have gathered after accounting for autocorrelation. The values presented in the table below represent the number of independent samples that the MCMC chains are effectively equivalent to. We obtain the following results:

β	γ	ι	δ
1302.206	1185.386	1284.587	1151.749

Table 4: ESS for 3 MCMC chains at initial parameters $(0.07, 0.03, 80, 0.15)$, $(0.06, 0.025, 85, 0.14)$ and $(0.08, 0.035, 75, 0.16)$.

The table above has an even distribution and large numbers suggesting our MCMC algorithm successfully draws independent samples from our posterior distribution for each of the parameters. The independence is importance to avoid autocorrelation between parameters and ensures our MCMC algorithm is mixing well.

Q2e): Recalling our data, the MLE from question 1 is $(\beta, \gamma, \iota, \delta) = (0.08574, 0.00692, 77.73077, 0.05)$ where we fixed $\delta = 0.05$. From our Bayesian approach, our posterior medians are $(\beta, \gamma, \iota, \delta) = (0.0708, 0.0269, 88.760, 0.155)$. We observe that β is estimated similar through the two methods, however γ is very different in the two approaches. ι and δ differ slightly too. Some strengths of using a Bayesian approach over MLE estimates are:

- Allows us to incorporate prior knowledge, using Bayesian inference allows for the inclusion of prior knowledge which is especially useful when we wish to include expert knowledge into our model. MLE methods do not allow us to do this and are more so a direct computational optimisation strategy.

- Instead of producing a single point for each parameter, like MLE, the Bayesian method allows us to produce a distribution for each parameter giving better insights into the shape of a parameter with different levels of confidence.
- Bayesian methods are useful for small sample sizes because we use a prior distribution, so when data is limited the inclusion of prior knowledge becomes very useful in producing good results. MLE methods require larger datasets to identify the general trend for the given dataset.

Some weaknesses to using a Bayesian approach over MLE estimates are:

- Bayesian methods are very computational expensive as we observed with the MCMC algorithms, where as MLE optimisation can be achieved much quicker and is less expensive.
- The choice for a prior distribution is subjective so Bayesian methods have a subjective component unlike MLE estimates which are concrete optimisers. This can introduce bias into our results if the prior distribution is chosen poorly.
- Bayesian inference is often harder to interpret for a non-statisician compared to MLE methods since they involve more complex probability concepts such as priors, posteriors, credible intervals.
- Although both methods can have convergence issues, MCMC algorithms are more prone to convergence issues since they require more careful tuning (e.g. burn-in, thinning, mixing).